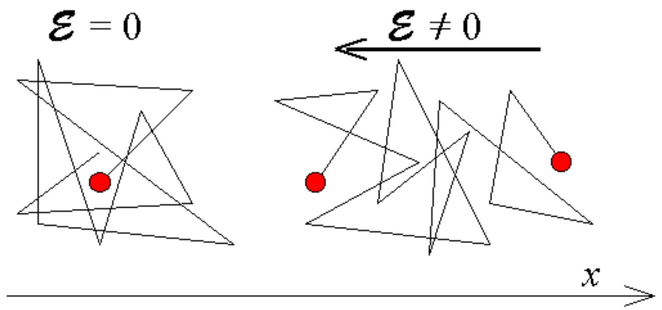
Fundamentals of Photovoltaics (EE336)

Revision for P-N Junction Diode

Carrier Transport

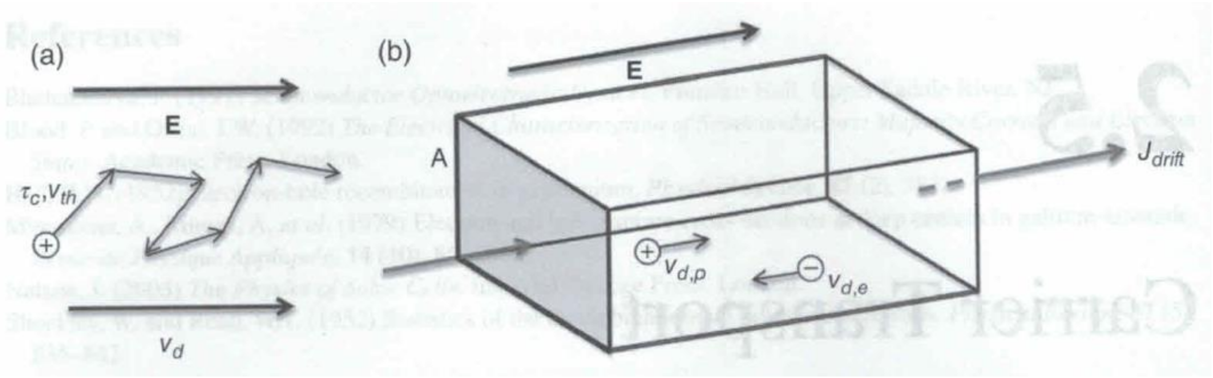
- Electrons in the conduction band and holes in the valence band are considered "free" carriers in the sense that they can move throughout the semiconductor lattice that makes up the crystal structure of the material. Each carrier as moving in a random direction at a certain velocity.
- The carrier moves in this random direction for a distance called the scattering length before colliding with a lattice atom. Once the collision takes place, the carrier moves away in a different random direction.
- The velocity of the carriers is determined by the temperature of the lattice. The thermal velocity is an average carrier velocity. Carriers have a thermal velocity that is normally distributed around this average thermal velocity. Therefore, some carriers having a greater velocity and some lower.
- The average thermal velocity v_{th} of the particle can be obtained by

$$v_{th} = \sqrt{\frac{3k_BT}{m}}$$



- **Although carriers in a semiconductor are in constant random motion, there is no *net* motion of carriers unless there is a concentration gradient or an electric field.**

Drift Current



In the absence of an electric field, carriers move a certain distance at a constant velocity in a random direction. However, in the presence of an electric field, superimposed on this random direction, and in the presence of thermal velocity, carriers move in a net direction.

Upon application of an electric field, carriers are accelerated along the filed direction between collisions. This will lead to a net carrier velocity component, which is called the drift velocity, v_d (cm/s). The drift velocity is related to the electric field by following equations:

For hole drift velocity; $v_{dp} = \frac{q\tau_{cp}}{m_p^*} E = \mu_p E$

For electron drift velocity; $v_{dn} = -\frac{q\tau_{cn}}{m_n^*} E = -\mu_p E$

where τ_c is the scattering time (mean time between the collision), m* is the effective mass and μ is the mobility

The subscript *n* and *p* represent electron and hole respectively

The drift current density J_{driff} is then found by considering the drift velocity of all available carriers.

$$J_{drift} = J_{n,drift} + J_{p,drift} = (qn\mu_n + qp\mu_p)E$$

The constant of proportionality relating current density and electric field is called the conductivity, σ (Scm^{-1}) and its inverse is called the resistivity, ρ (Ω.cm)

$$\sigma = \frac{1}{\rho} = qn\mu_n + qp\mu_p$$

The mobility of a semiconductor (and ultimately its conductivity) depends on the effective mass of the material, with low effective mass materials having higher mobility. As well, the scattering time factor leads to the dependence of the mobility on the temperature, doping, impurities and defects.

For many common semiconductors (e.g., Si and GaAs), there are often empirical relationships that can be used to find mobility dependence on doping and temperature.

The mobility decreases with higher doping concentration (impurities).

The mobility decreases with increasing temperature in a lattice scattering dominated region whereas the mobility increases with increasing temperature in an impurity scattering dominated region.

Diffusion Current

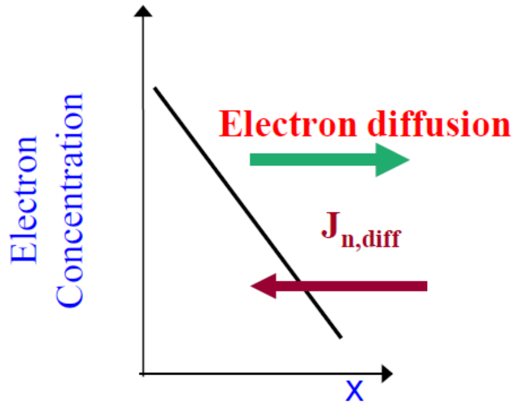
The diffusion current arises due to a concentration gradient of free carriers inside a semiconductor. As a result of their random thermal motion, carriers will always tend to flow from regions of high to low concentration.

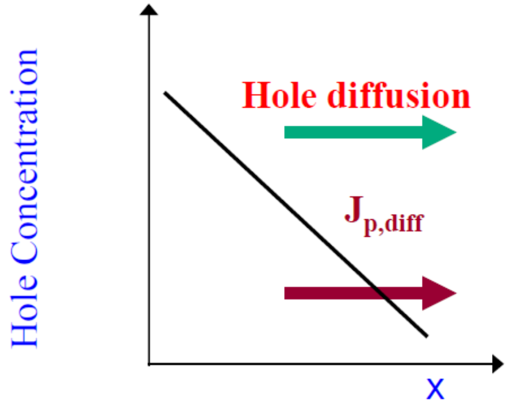
According to Fick's First Law of Diffusion, under a steady state condition, particle flux is proportional to the concentration gradient, where the proportionality constant is called the diffusion coefficient. If the particles are free charge carriers, the flux will give rise to a current.

- The electron diffusion current density, $J_{n,diff} = qD_n \frac{dn}{dx}$
- The hole diffusion current density, $J_{p,diff} = -qD_p \frac{dp}{dx}$

where D_n and D_p are diffusion coefficient of electron and hole.

The total diffusion current is, $J_{diff} = J_{n,diff} + J_{p,diff} = qD_n \frac{dn}{dx} - qD_p \frac{dp}{dx}$





The diffusion coefficient and mobility are not independent material properties. By considering the mechanics of particles in a solid, it can be shown that the diffusion coefficient and mobility are related by Einstein relationship.

$$D_n = \frac{kT}{q}\,\mu_n$$

$$D_p = \frac{kT}{q}\,\mu_p$$

Total Current

The total carrier current is then the linear sum of drift and diffusion components. Breaking these down by particle type, we obtain:

$$J_n = J_{n,drift} + J_{n,diff} = qn\mu_n E + qD_n\frac{dn}{dx}$$

$$J_p = J_{p,drift} + J_{p,diff} = qp\mu_p E - qD_p\frac{dp}{dx}$$

The total current is the sum of each type of carrier current:

$$J_{total} = J_n + J_p$$

Continuity Equations

The carrier processes discussed in the previous sections (current flow, recombination and generation) can be related to one another through the continuity equations.

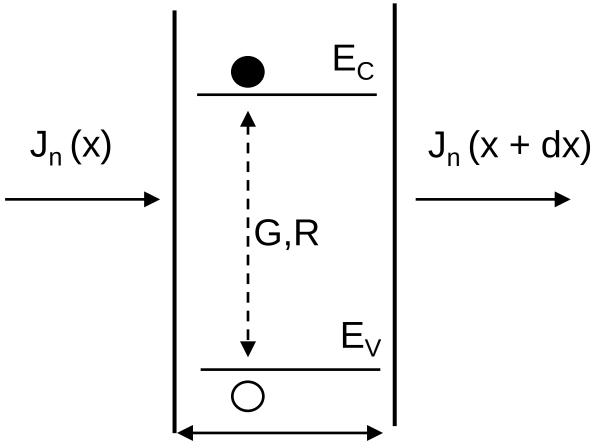
Consider the example in below figure for electron flow into a one-dimensional region of a semiconductor.

The time rate of change of electrons in the slice *dx* can be express as

$$\frac{dn}{dt} = \frac{1}{q}\frac{dJ_n}{dx} + (G_n - U_n)$$

where G_n is the electron generation rate and U_n is the electron recombination rate.

The analysis can be applied to both electrons and holes and extends to three dimensions.

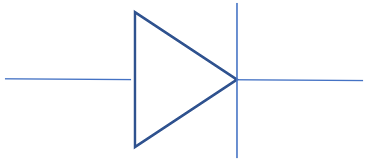


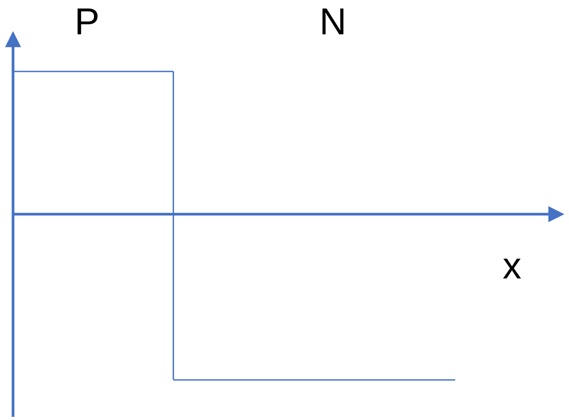
P-N junction

P-N junction is a semiconductor with part of it doped p-type with acceptor impurities and the other part doped with n-type donor impurities.

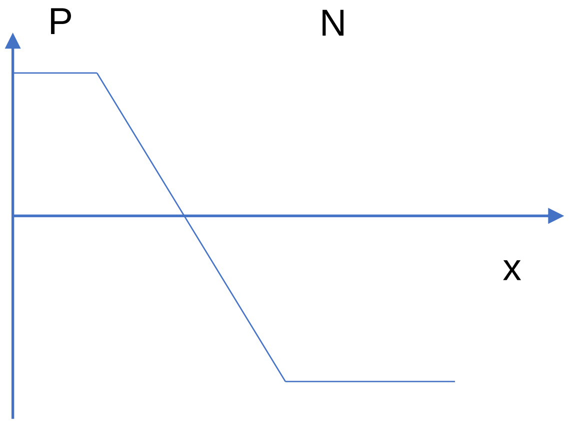
An abrupt p-n junction or a step junction is one where the dopant concentration changes abruptly (actually over a small distance).

A diffused pn junction is one where the transition occurs over a lager distance.





abrupt/step junction



diffused junction

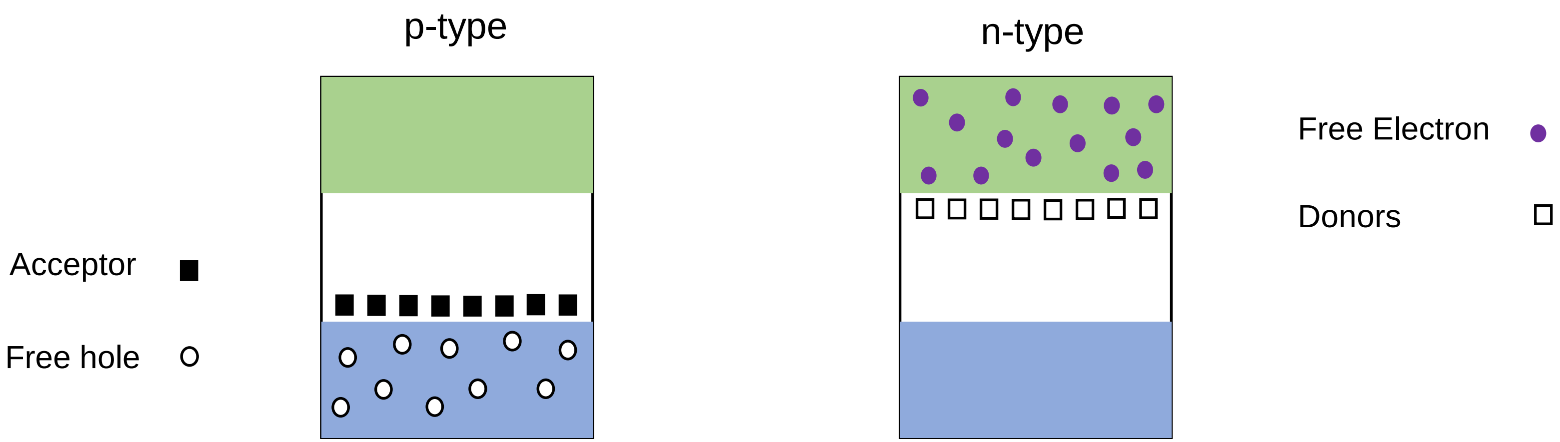
P-N Junction in Equilibrium

Solar cell is a p-n junction diode that operates under illumination, i.e., non-equilibrium condition in reality. However, before studying the non-equilibrium condition, we should understand the p-n junction in equilibrium condition first. The equilibrium condition implies:

- No application of any external voltage
- No light shining (dark conditions)
- Uniform temperature through out the semiconductor
- No externally applied electrical or magnetic fields

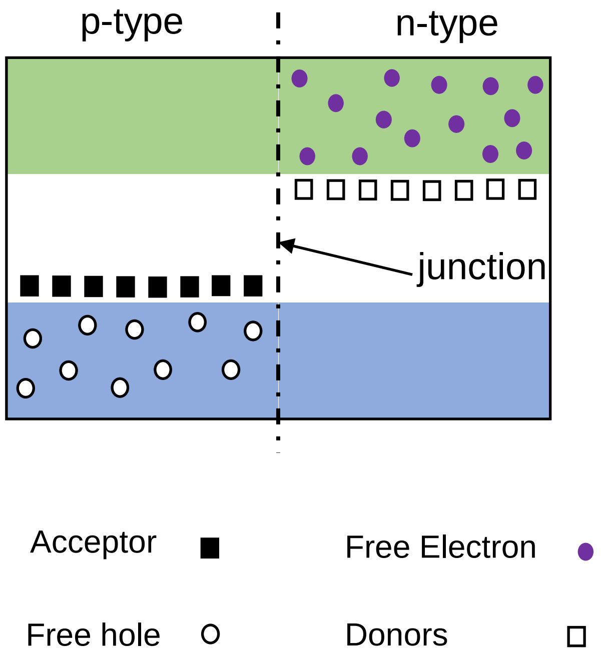
In short, the p-n junction is maintained at a constant temperature with NO external disturbance of any kind.

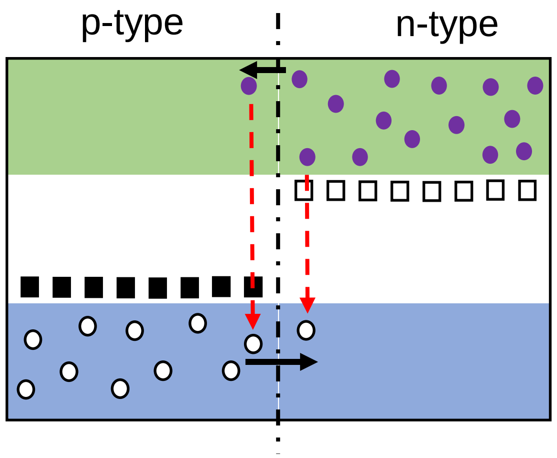
To understand how a p-n junction diode works, begin by imagining two separate bits of semiconductor, one n-type, the other p-type.



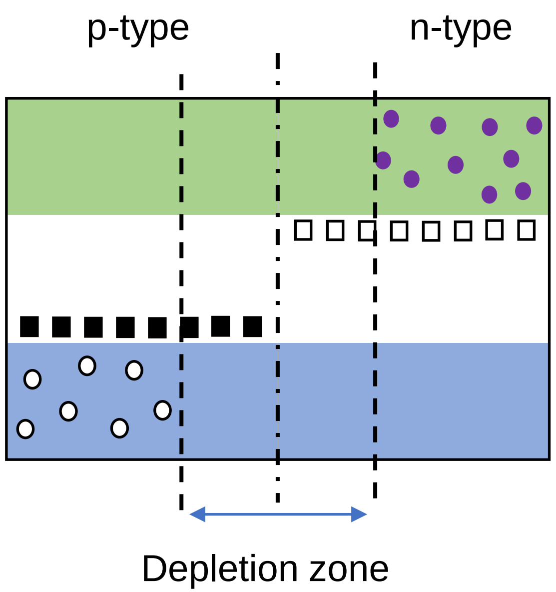
Bring them together and join them to make one piece of semiconductor which is doped differently on either side of the junction.

Free electrons on the n-side and free holes on the p-side can initially wander across the junction. When a free electron meets a free hole, it can "drop into it". So far as charge movements are concerned, this means the hole and electron cancel each other and vanish.

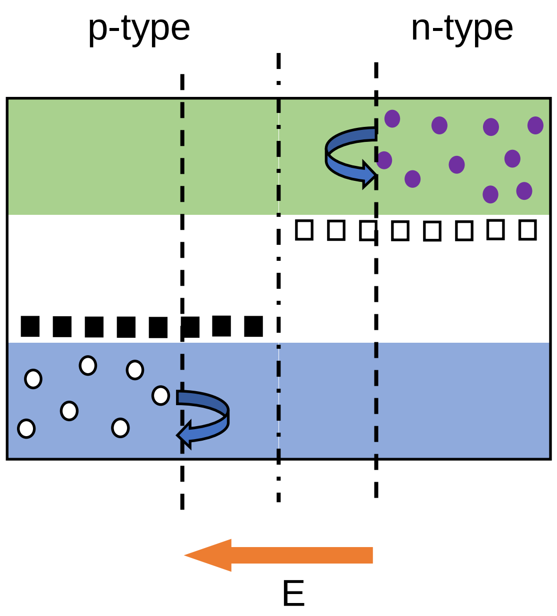




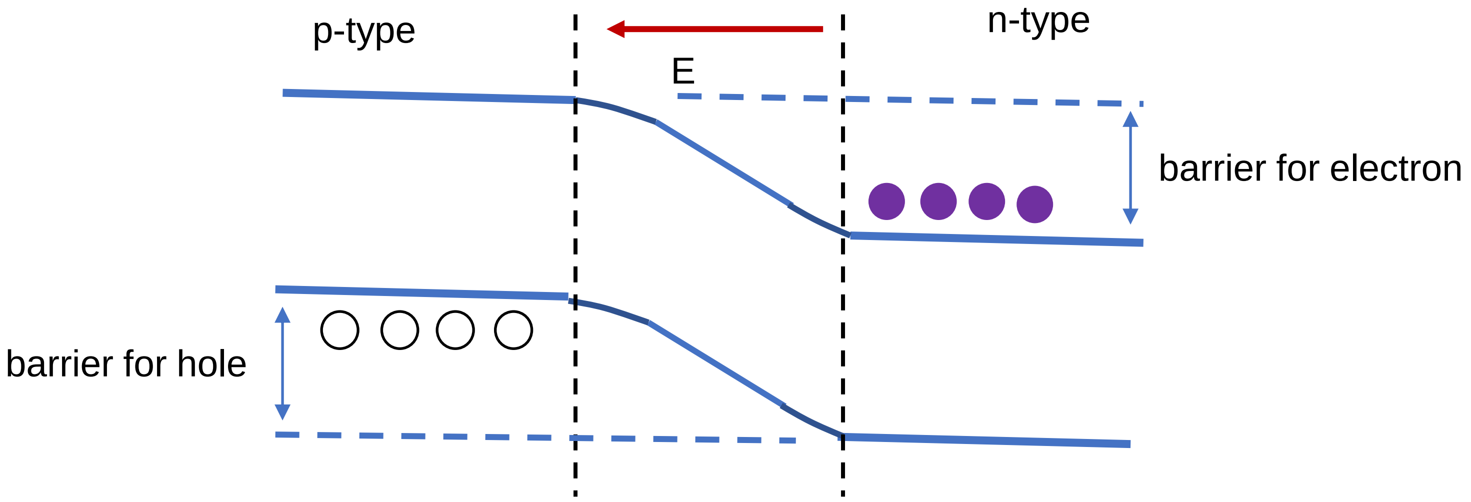
As a result, the free electrons and holes near the junction tend to eat each other, producing a region depleted of any moving charges. This creates what is called the depletion zone or space charge zone.



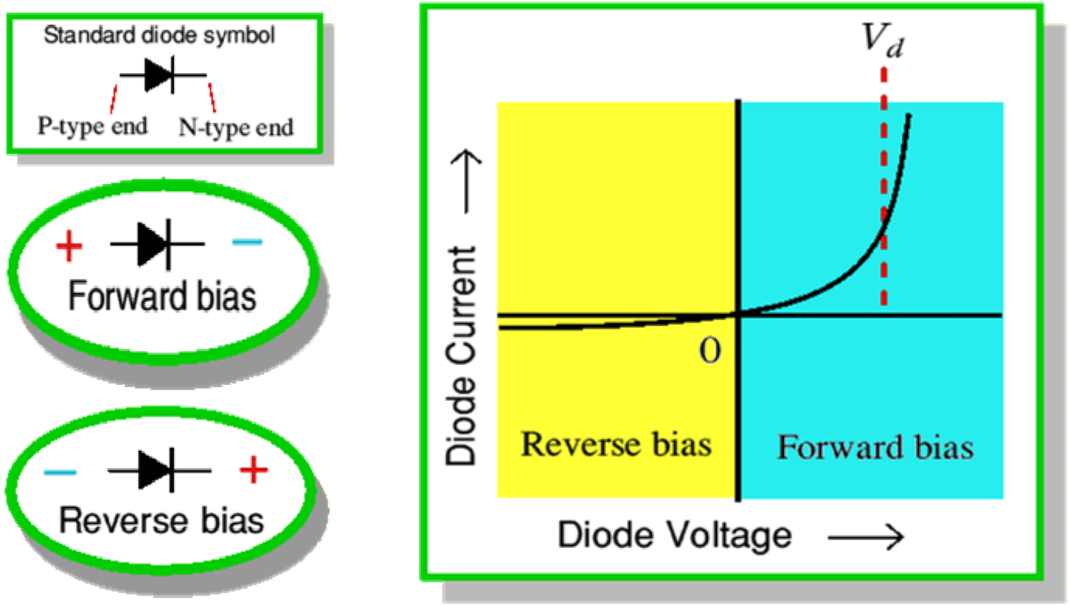
Now, any free charge which wanders into the depletion zone finds itself in a region with no other free charges. Locally it sees a lot of positive charges (the donor atoms) on the n-type side and a lot of negative charges (the acceptor atoms) on the p-type side. These exert a force on the free charge, driving it back to its “own side” of the junction away from the depletion zone.



- The acceptor and donor atoms are “nailed down” in the solid and cannot move around.
- However, the negative charge of the acceptor’s extra electron and positive of the donor’s extra proton (exposed by it’s missing electron) tend to keep the depletion zone swept clean of free charges once the zone has formed.
- A free charge now requires some extra energy to overcome the forces from the ionized donor/acceptor atoms to be able to cross the zone. The junction therefore acts like a barrier, blocking any charge flow (current) across the barrier.
- Usually, we represent his barrier by bending the conduction and valence bands as they cross the depletion zone.
- Now we can imagine, the electrons having to “get uphill” to move from the n-type side to the p-type side.
- The holes behave a bit like balloons bobbing up against a ceiling. On this kind of diagram, you require energy to “pull them down” before they can move from the p-side to the n-type side.
- The energy required by the free holes and electrons (to move to other side) can be supplied by a suitable voltage applied between the two ends of the p-n junction diode.



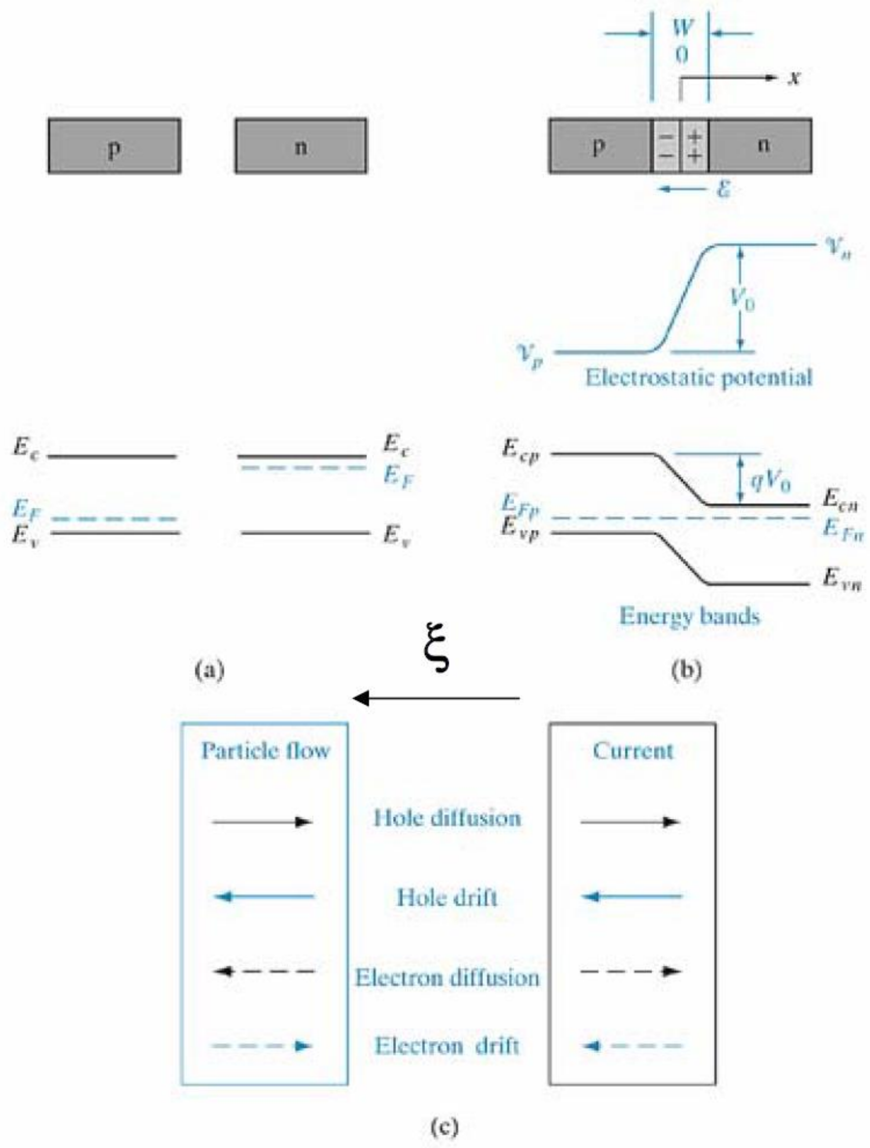
- Notice that this voltage must be supplied the correct way, around, this pushes the charges over the barrier (p-n junction forward bias)
- However, applying the voltage the “wrong” way around makes things worse by pulling what free charges there are away from the junction (p-n junction is in reverse bias)!
- This is why p-n junction (or diode) conducts in one direction but not the other.



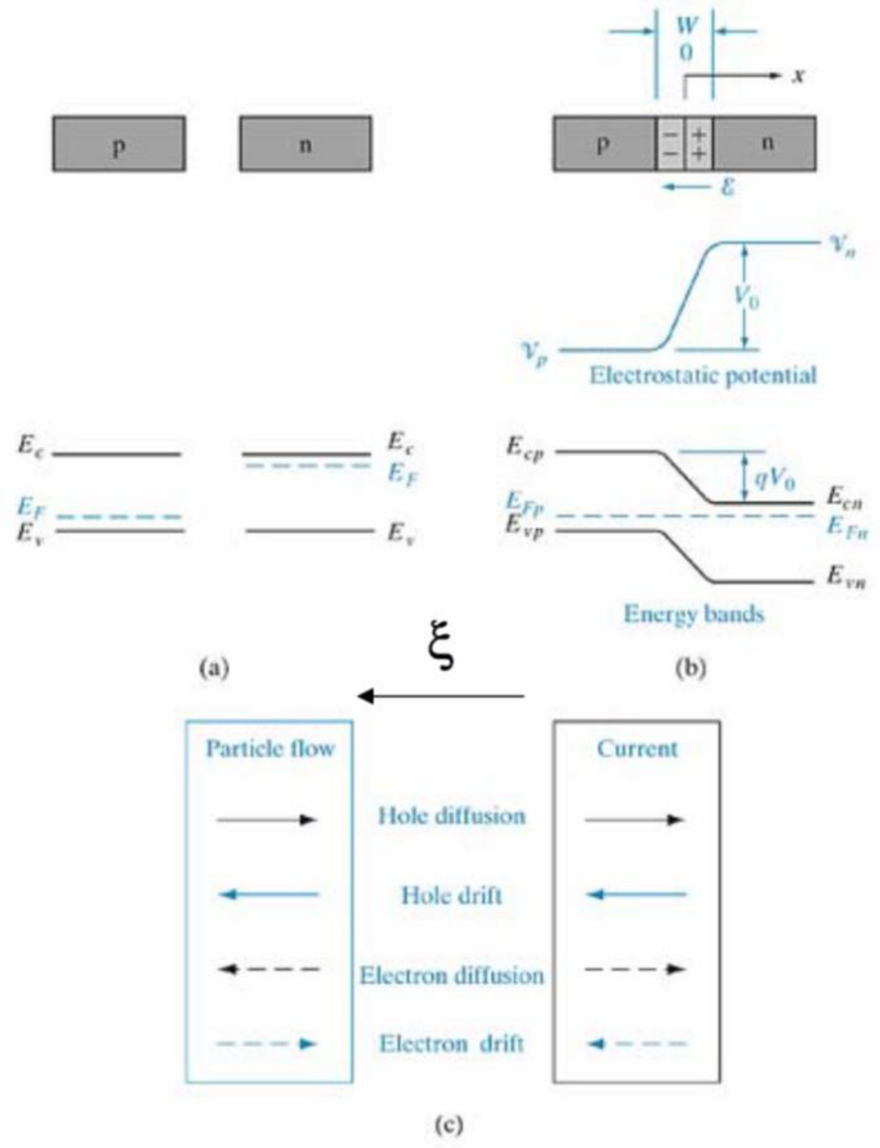
Forward bias - all current, almost no volts.
Reverse bias - all volts, almost no current.

Upon the formation of p-n junction

- Hole diffuse from p-side to n-side and electron diffuse from n-side to p-side.
- The diffusion of holes and electrons give rise to corresponding diffusion currents, J_p(diff) and J_n(diff), respectively.
- Both J_p(diff) and J_n(diff) flow from p-side to n-side.
- The diffusion current cannot build up indefinitely because of an opposing electric field that is created at the junction.



- Electrons diffusion from n to p leave behind uncompensated (ionized) donor ions N_d^+ in the n-material.
- Holes diffusion from p to n leave behind uncompensated (ionized) acceptor ions N_a^- in the p-material.
- The resulting electric field (E) is directed from n-side to p-side.
- The electric field E is in a direction opposite to that of diffusion current of both electrons and holes – lowers diffusion current.
- The electric field E leads to drift currents ($J_{driff} = \sigma E$)
- J_p(drift) and J_n(drift) flow from n-side to p-side, in the direction of the electric field E.



At equilibrium, there is no net current flow across the p-n junction.

This means the drift current due to electric field E must exactly balance the diffusion current.

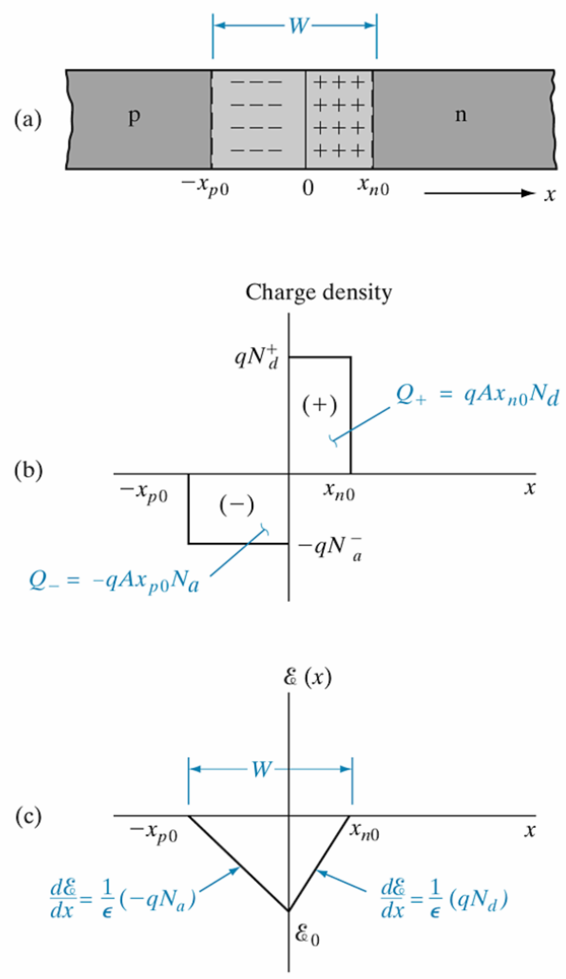
Also, since there is no net build up of either electrons or holes on either side as a function of time, the drift and diffusion currents must cancel for each type of carrier.

$J_p = J_p(\text{drift}) + J_p(\text{diff}) = 0$

$J_n = J_n(\text{drift}) + J_n(\text{diff}) = 0$

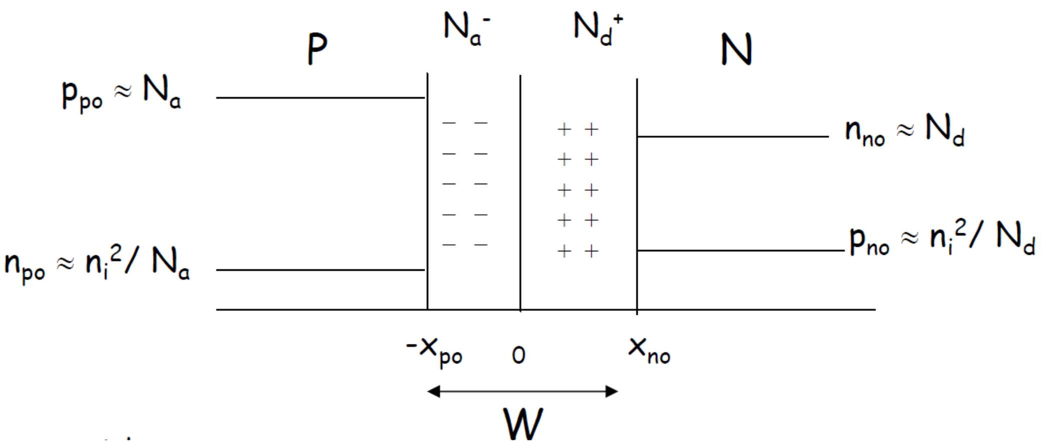
The total current density $J = J_p + J_n = 0$, at thermal equilibrium.

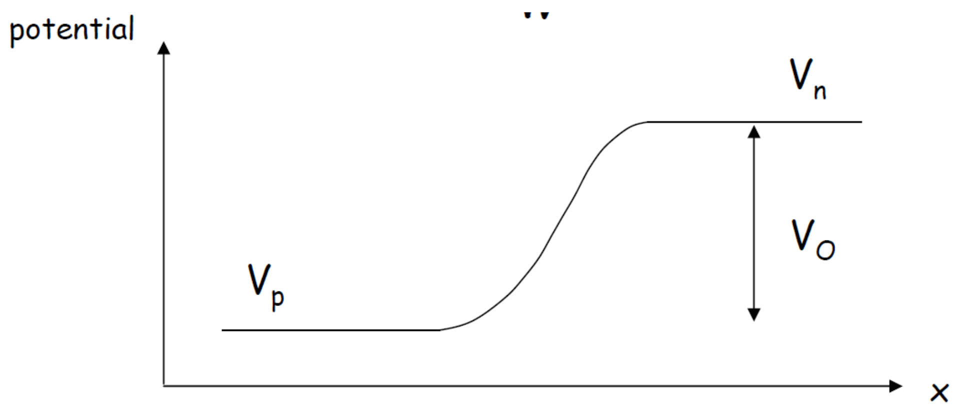
- The electric field E thus would have to build up to the point where the net current *J* is zero at equilibrium.
- This electric field E appears across a region W about the junction, and there is an equilibrium potential difference V_o across W.
- The electric field is zero in the neutral regions of the semiconductor outside W – implies a constant potential V_n in the neural n-material and a constant potential V_p in the neutral p-material.
- There exists thus, a potential difference $V_o = V_n - V_p$ over a region of width W.
- The region W is called the depletion region (also called the transition or space-charge region) and the potential difference V_o is called the contact potential (also called build-in potential).
- Note that the contact potential appearing across the depletion region is a built-in barrier and not an external potential. Thus, the contact potential cannot be measured by placing a voltmeter across a p-n junction.



Space charge and electric field distribution within the transition region of a p-n junction with *Nd* > *Na*: (a) the transition region, with x = 0 defined at the metallurgical junction; (b) charge density within the transition region, neglecting the free carriers; (c) the electric field distribution, where the reference direction for ξ is arbitrarily taken as the +x-direction. 27

P-N Junction in equilibrium – Contact Potential





The contact potential can be calculated by

$$V_o = \frac{kT}{q}\ln\left(\frac{p_{p0}}{p_{n0}}\right) = \frac{kT}{q}\ln\left(\frac{N_aN_d}{n_i^2}\right)$$

At equilibrium, $p_{po}n_{po} = n_i^2 = n_{no}p_{no}$

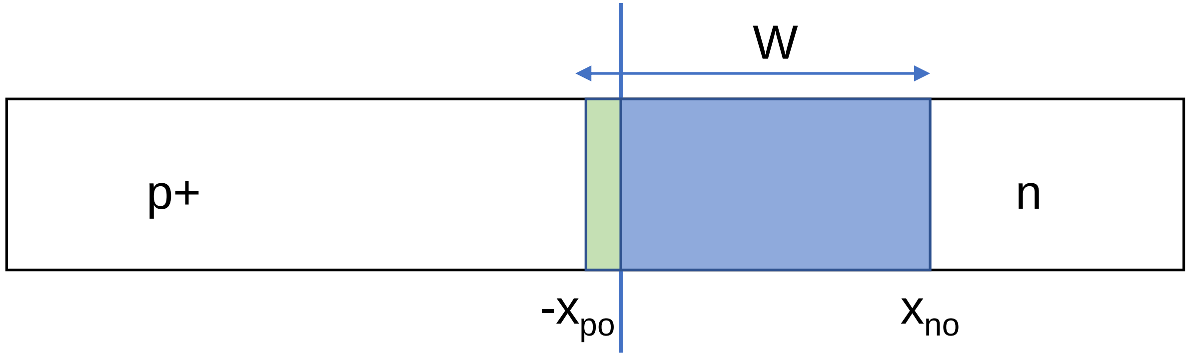
Thus,

$$\frac{p_{p0}}{p_{no}} = \frac{n_{no}}{n_{p0}} = e^{\frac{qV_0}{kT}}$$

P-N junction in equilibrium – Space Charge

- In the depletion region, electrons and holes are in transit, i.e., some electrons diffusing from n to p and some electrons swept by the electric field from p to n (vice-versa for holes).
- The depletion region can be considered to consist of only charges due to uncompensated donor and acceptor ions.
- There is charge neutrality outside the depletion region (zero net charge in the neutral semiconductor)
- The depletion region W will penetrate further to the more lightly doped side, e.g. if $N_a > N_d$ then $x_{po} < x_{no}$ and vice-versa
- In many instances, p-n junctions with one side much heavier doped than the other are encountered. These are called p^+n. In such junction the depletion region extends almost entirely into the more lightly doped side.

- $x_{n0} = \frac{WN_a}{N_a+N_d}$
- $x_{p0} = \frac{WN_d}{N_a+N_d}$



The plot of E (x) vs x has two slopes, positive on n-side and negative on p-side. The maximum E occurs at x =0, the metallurgical junction.

The peak electric field is

$$E_0 = -\frac{q}{\varepsilon} N_a x_{p0} = -\frac{q}{\varepsilon} N_d x_{n0}$$

The contact potential can also be obtained by integrating E(x) over x. Thus

$$V_0 = -\frac{1}{2} E_0 W$$

$$W = \left[\frac{2\, \varepsilon V_0}{q}\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2}$$

P-N junction in equilibrium – Summary

$$V_o = \frac{kT}{q}\ln\left(\frac{p_{p0}}{p_{n0}}\right) = \frac{kT}{q}\ln\left(\frac{N_a N_d}{n_i^2}\right)$$

$$\frac{p_{p0}}{p_{no}} = \frac{n_{no}}{n_{p0}} = e^{\frac{qV_0}{kT}}$$

$$x_{n0} = \frac{WN_a}{N_a + N_d} \quad , \qquad x_{p0} = \frac{WN_d}{N_a + N_d}$$

$$E_0 = -\frac{q}{\varepsilon}N_a x_{p0} = -\frac{q}{\varepsilon}N_d x_{n0}$$

$$W = \left[\frac{2\,\varepsilon V_0}{q}\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2}$$

P-N junction under bias

Built-in potential in the depletion region

- is lowered under FB ($V = V_f$), it decreases to ($V_o – V_f$)
- Is increased under RB ($V = -V_r$), it increases to ($V_o + V_r$)

Electric field in the depletion region

- under FB, E decreases (external field opposes built-in field)
- under RB, E increases (external field aids built-in field)

Depletion region with W

- under FB, W decreases
- under RB, W increases

$$W_{FB} = \left[\frac{2\,\varepsilon(V_0 - V_f)}{q}\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2} \qquad W_{RB} = \left[\frac{2\,\varepsilon(V_0 + V_r)}{q}\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2}$$

P-N junction under bias

<u>Energy band separation</u>

- under FB, $E_{cp} - E_{cn} = q(V_o - V_f)$ – decreases
- under RB, $E_{cp} - E_{cn} = q(V_o + V_r)$ – increases

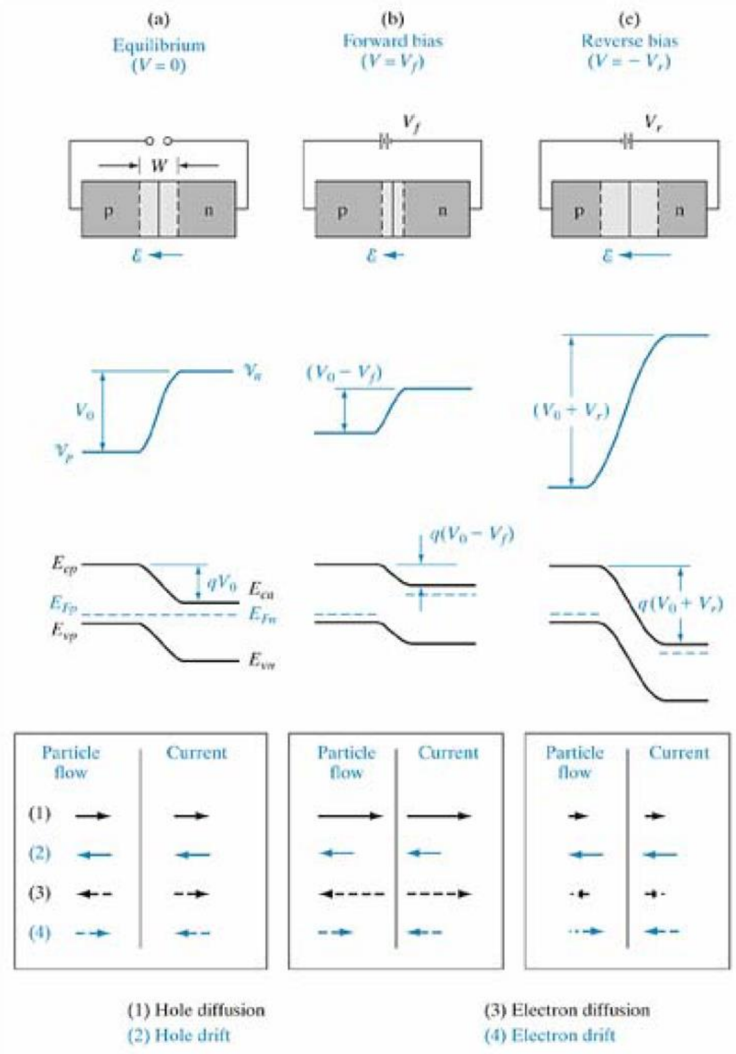
Bands are separated less in FB and more in RB compared to thermal equilibrium condition.

Fermi levels E_{Fp} and E_{Fn} are no longer aligned (applied V results in a non-equilibrium condition). Their separation is numerically equal to the applied voltage, V_f for FB and V_r for RB.

In summary

under FB, built-in potential $\downarrow$, E $\downarrow$, W $\downarrow$, energy barrier $\downarrow$

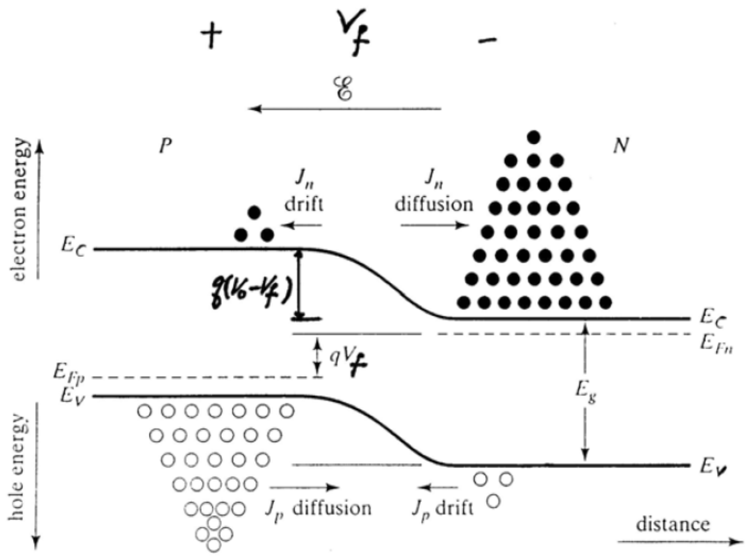
under RB, built-in potential $\uparrow$, E $\uparrow$, W $\uparrow$, energy barrier $\uparrow$



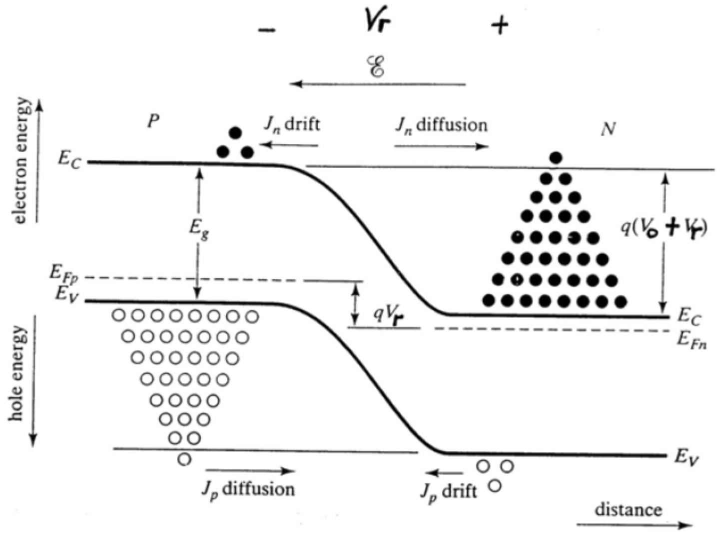
Effects of a bias at a p-n junction; transition region width and electric field, electrostatic potential, energy band diagram, and particle flow and current directions within W for (a) equilibrium, (b) forward bias, and (c) reverse bias.

P-N junction under bias – Diffusion current

- With zero bias, there is no diffusion current. (strictly speaking, there is small amount of diffusion and drift current under equilibrium but it is very small and can be considered as zero)
- Under FB, the potential barrier for both electrons and holes is reduced to $q(V_o - V_f)$
- Under RB, the potential barrier for both electrons and holes is increased to $q(V_o + V_r)$



Forward Bias



Reverse Bias

- Under FB, electrons in the n-side conduction band (CB) have sufficient energy to diffuse from n to p over reduced potential barrier (same for holes in valence band (VB))
- Thus, the electron (and hole) diffusion current increases sharply with FB.
- Under RB, the increased barrier prevents electrons in the n-side CB to diffuse to p-side (same for holes in VB)
- Thus, electrons and hole diffusion currents, J_n(diff) and J_p(diff)
    - ✓ increase sharply under FB
    - ✓ decrease sharply under RB
- Although the applied voltage V affects the built-in field E, J_n(drift) and J_p(drift) are in fact not affected by the strength of E and hence the applied bias V.
- The carriers that are participating in the drift process are derived from the minority carriers in the neutral p and n regions just outside W.
- Minority electrons (holes) at the p-side (n-side) that wander into the depletion region will be swept by the E field across W, giving rise to J_n(drift) (J_p(drift))
- J_n(drift) and J_p(drift) are determined by the number of minority carriers entering W per sec, NOT by the electric field strength E in W.
- Thus, J_n(drift) and J_p(drift) are insensitive to the applied bias V, and are only determined by the concentration of minority carriers in the neutral p and n regions just outside the depletion region W.

P-N junction under bias – Carrier Concentration

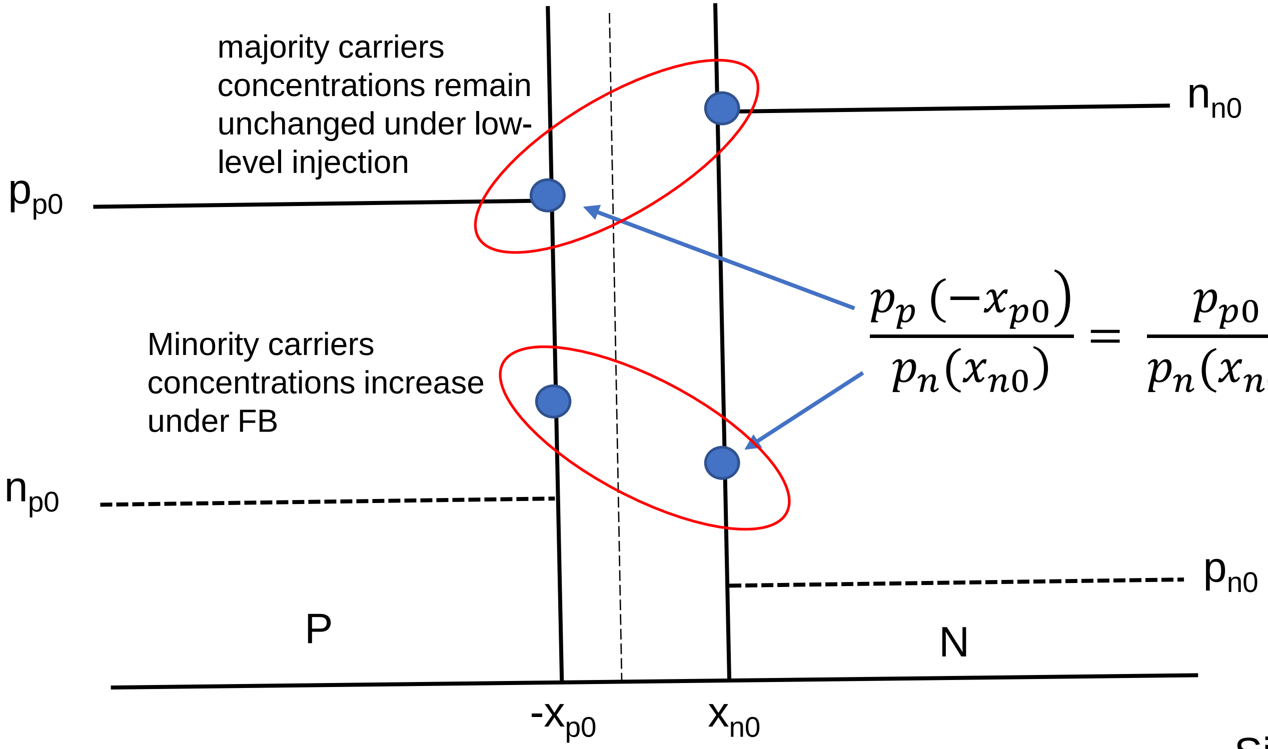
- At thermal equilibrium, the concentration of the minority carriers in the neutral p and n regions just outside W are given by $n_{po} = n_i^2 / N_a$ (p-side) and $p_{no} = n_i^2 / N_d$ (n-side). These equilibrium concentrations are maintained because the number of minority carriers lost per sec due to the drift process is just exactly balanced by the number of minority carrier being brought back by the diffusion process. (J_n(drift) = J_n(diff), J_p(drift) = J_p(diff))
- Under FB where diffusion of carriers is enhanced whereas drift current remains unchanged (J_n(drift) < J_n(diff), J_p(drift) < J_p(diff)). The minority carrier concentrations in the neutral p and n regions outside W will be built-up above the equilibrium values due to the net diffusion of carriers. This is called <u>minority carrier injection</u>.
- Under RB where diffusion of carriers is inhibited whereas drift current remains unchanged (J_n(drift) > J_n(diff), J_p(drift) > J_p(diff)). The minority carrier concentrations in the neutral p and n regions outside W will drop below the equilibrium values due to the net drift of carriers. This is called <u>minority carrier extraction</u>.

- When the p-n junction is in equilibrium, the majority carrier concentration on p-side p_{po} is related to the minority carrier on the n-side p_{no} through the equilibrium contact potential V_o. The same for n-side.

$$\frac{p_{p0}}{p_{no}} = \frac{n_{no}}{n_{p0}} = e^{\frac{qV_0}{kT}}$$

- In FB, hole will diffuse from p-side to n-side and electrons will diffuse from n-side to p-side. This would mean there is an increase in the minority carrier concentration both on the (neutral) p-side and n-side.
- Net hole diffusion from p to n-side brings excess minority holes $p_{n(ex)}$ to the n-side.
- Net electron diffusion from n to p-side brings excess minority electrons $n_{p(ex)}$ to the p-side.
- But the charge neutrality in the neutral p and n regions is still preserved because the battery will provide the necessary excess majority electrons $n_{n(ex)}$ to the n-side, and excess majority holes $p_{p(ex)}$ to the p-side.
- Assume that all the excess carriers in each side are much smaller compared to their respective majority carrier concentration (low-level injection)
- For low-level injection, the injected minority carrier do not upset the equilibrium majority carrier densities on both the p and n regions.

Forward Bias, Low Level Injection

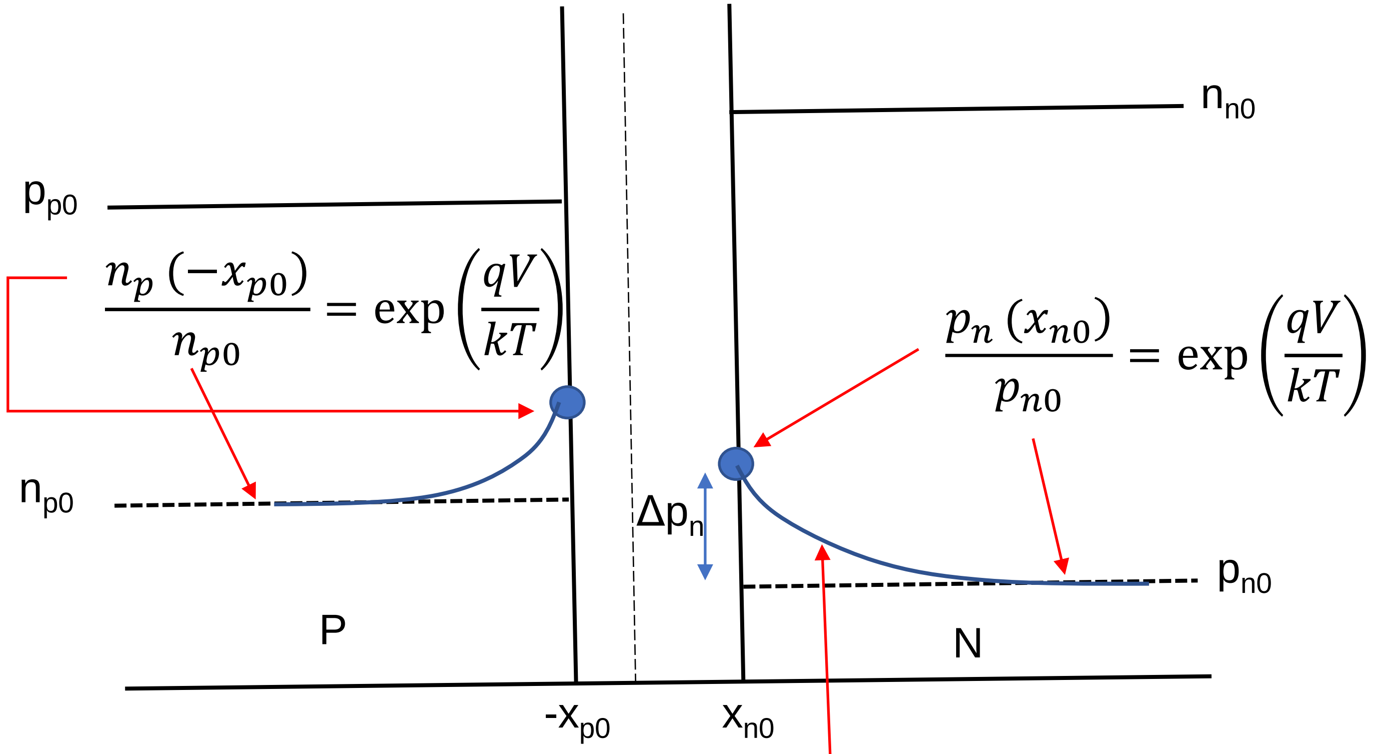


$$\frac{p_p(-x_{p0})}{p_n(x_{n0})} = \frac{p_{p0}}{p_n(x_{n0})} = \exp\left(q\frac{(V_0 - V)}{kT}\right)$$

Similarly, $$\frac{n_n(x_{n0})}{n_p(-x_{p0})} = \frac{n_{n0}}{n_p(-x_{p0})} = \exp\left(q\frac{(V_0 - V)}{kT}\right)$$

New minority hole concentration at x_{n0} is denoted as $p_n(x_{n0})$ and new minority electron concentration at $-x_{p0}$ is denoted as $n_p(-x_{p0})$

Forward Bias, Low Level Injection



$$\delta p\,(x_n) = \Delta p_n e^{\frac{-x_n}{L_p}}$$

$$\Delta p_n = p_n(x_{n0}) - p_{n0} = p_{n0}\left(e^{\frac{qV}{kT}} - 1\right)$$

$$\Delta n_p = n_p(-x_{p0}) - n_{p0} = n_{p0}\left(e^{\frac{qV}{kT}} - 1\right)$$

P-N junction under bias – Summary

- under FB, built-in potential $\downarrow$, E $\downarrow$, W $\downarrow$, energy barrier $\downarrow$
- under RB, built-in potential $\uparrow$, E $\uparrow$, W $\uparrow$, energy barrier $\uparrow$

Diffusion Currents J_n(diff) and J_p(diff)

- increase sharply under FB
- decrease sharply under RB

Drift Currents J_n(drift) and J_p(drift) ---- DO NOT DEPEND ON BIAS

Carrier Concentration

FB ---- minority carrier injection

RB ---- minority carrier extraction

Diode Current

We've known the information about minority carrier concentrations at both p and n-side under forward bias and low-level injection. Now we ready to formulate the diode current.

Consider the total current at a particular point (e.g. x_{n0}). It consists of two types of carriers: majority electrons ($I_n(x_{n0})$) and minority holes ($I_p(x_{n0})$).

$I(x_{n0}) = I_n(x_{n0}) + I_p(x_{n0})$

Let's consider minority hole current $I_p(x_{n0})$. It has diffusion and drift components. However, the drift component can be neglected because holes are minority carriers meaning that its concentration is low and the electric field is ~0 in the neutral region (most of the applied bias drops across W and only negligible voltage drops across the neutral p and n regions).

Therefore, $I(x_{n0}) = I_n(x_{n0}) + I_p(x_{n0})_{diff}$

To find the majority electron current $I_n(x_{n0})$ at x_{n0}, the edge of W on the n-side, we have to look at the minority electron current at the other edge of W, at $-x_{p0}$, on the p-side. This current is $I_n(-x_{p0})$, which is nothing but electron diffusion current at $-x_{p0}$.

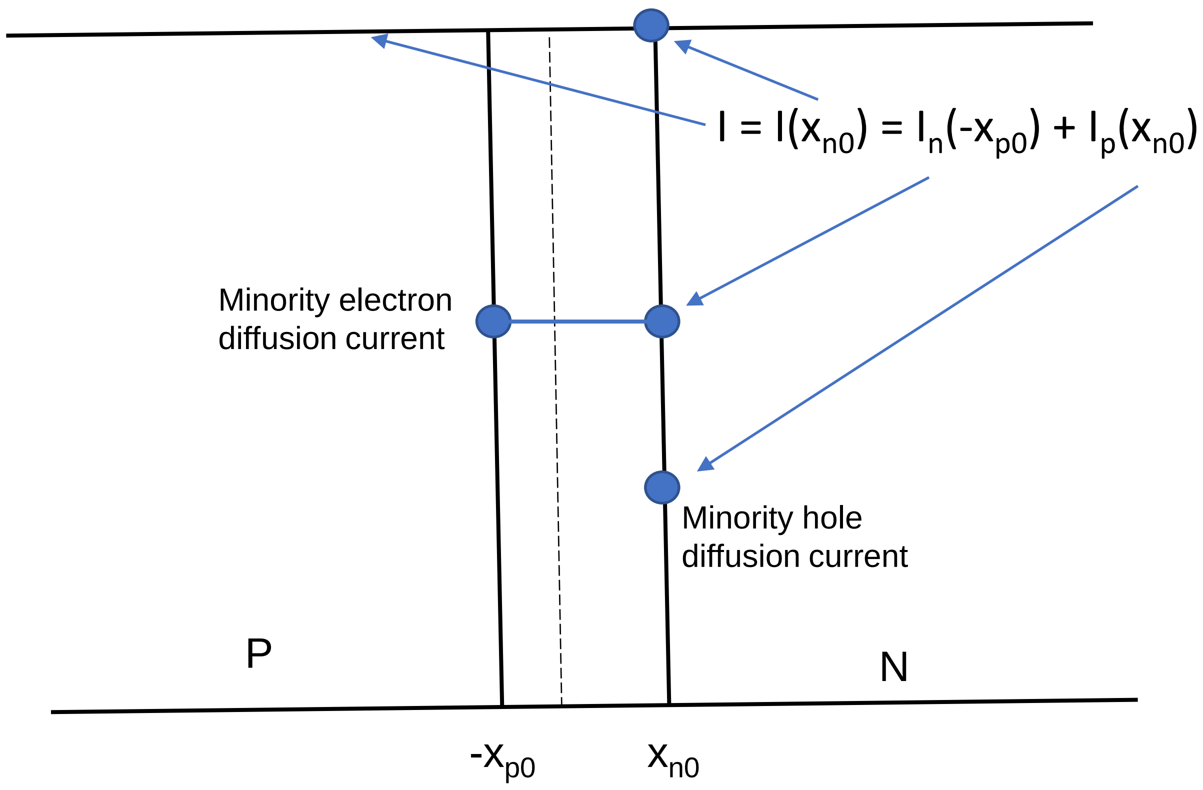
Recall that the depletion region W extends from $-x_{p0}$ to x_{n0} and in the depletion region, there is no recombination or generation of electrons. This means that the minority electron current $I_n(-x_{p0})$ must be equal to the majority electron current $I_n(x_{n0})$.

$I_n(-x_{p0})$ is a minority electron current with both diffusion and drift components but we can neglect its drift component simply because the electric field is ~0 out the W.

Thus, the total current I at x= x_{n0} is

$$I(x_{n0}) = I_n(x_{n0}) + I_p(x_{n0})$$
$$= I_n(-x_{p0}) + I_p(x_{n0})$$
$$\approx I_n(-x_{p0})_{diff} + I_p(x_{n0})_{diff}$$

This current is, simply, a sum of the two minority carrier diffusion currents at the edges of depletion region W.



The minority diffusion currents at $-x_{p0}$ and x_{n0} can be calculated using diffusion current equation ($I_n = qAD_n \frac{dn}{dx}$, $I_p = -\,qAD_p \frac{dp}{dx}$) and setting x = 0.

$$I\,(x_{n0}) = \frac{qAD_n}{L_n}\,\Delta n_p + \frac{qAD_p}{L_p}\,\Delta p_n$$
$$= qA\,\left[\frac{D_n}{L_n}n_{p0} + \frac{D_p}{L_p}p_{n0}\right]\left[e^{\frac{qV}{kT}} - 1\right] = \,I_0\left[e^{\frac{qV}{kT}} - 1\right]$$

where I_0 is called the reverse saturation or generation current.

$$I_0 = qA\left[\frac{D_n}{L_n}n_{p0} + \frac{D_p}{L_p}p_{n0}\right]$$

Since the current flow must be the same at every point in the p-n junction, the total current flowing at x = x_{n0} ($I(x_{n0})$) must also be the current flowing through all x. The diode current equation

$$I \;=\; I_0\left[e^{\frac{qV}{kT}} - 1\right]$$

is called the ideal diode equation that depicts the current of the p-n junction as a function of the bias voltage V. The equation is valid for both FB and RB.

The above equation is for ideal diode. In reality, the diode current can be deviated from the ideal diode current because of the recombination in the bulk and surface of semiconductor. The equation for non-ideal case is

$$I = I_0\left(e^{\frac{qV}{nkT}} - 1\right)$$

where n is the ideality factor, a number between 1 and 2 which typically increases as the current decreases.